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A few logs suffice to build (almost) all trees (I)

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Abstract

A phylogenetic tree, also called an “evolutionary tree,” is a leaf-labeled tree which represents the evolutionary history for a set of species, and the construction of such trees is a fundamental problem in biology. Here we address the issue of how many sequence sites are required in order to recover the tree with high probability when the sites evolve under standard Markov-style i.i.d. mutation models. We provide analytic upper and lower bounds for the required sequence length, by developing a new polynomial time algorithm. In particular, we show when the mutation probabilities are bounded the required sequence length can grow surprisingly slowly (a power of log n) in the number n of sequences, for almost all trees.

Real-World Impact & Application

Scalable Tree Inference with Short Genetic Sequences: Erdos et al. proved a landmark mathematical theorem in computational biology: accurate phylogenetic trees can be reconstructed even when DNA sequence length is only logarithmic in the number of taxa. This theoretical guarantee provided the mathematical assurance needed for large-scale automated biodiversity mapping and phylogenomic pipelines.